Synthetic data quality: dilution and fragmentation

Matched Qwen3-4B BABILong synthetic datasets; 27,949 examples per method

Method	Dilution and relevance			Fragmentation	
	No-support ↓	Query focus ↑	Support recall ↑	Joint exercise ↑	Joint case ↑
Vanilla Cartridges	83.78	1.77	52.31	0.00	0.00
Question-guided (ours)	**46.69**	**98.17**	**91.06**	**23.66**	**78.72**
Δ guided − agnostic (pp)	-37.09	+96.41	+38.75	+23.66	+78.72

Source diagnostics n=993; query focus n=1000; cross-chunk diagnostics n=296. These are evidence-exposure diagnostics, not answer accuracy.